



Fig. 11: Laser printer

eye surgery, treatment of stones and beauty therapies. Defence applications include precision-guided munitions, secure communication, high-energy lethal weapons and so on. Some common applications are briefly discussed below.

Computer applications. Lasers are widely used in reproduction of music stored in compact optical disks (CDs). These lasers are tiny inexpensive semiconductor lasers, buried deep inside compact disk players, which can only be observed if you disassemble the player.

Laser printers (Fig. 11) and scanners are very common these days. These can be connected individually to computers or accessed by a group of workers through a network.

Fibre-optic communications. Fibre-optics has become the dominant medium for long-distance telecommunications around the world. Optical fibres in submarine cables carry telephone calls under the Atlantic and Pacific oceans.

Materials processing. Materials processing is a broad field including cutting, welding, drilling and otherwise modifying industrial materials, including both metals and non-metals (see Fig. 12).

Medical applications. Physicians were not far behind engineers when it came to using lasers. Dermatologists and ophthalmologists were the first to use lasers. However, it was not long before other specialists started working with lasers. Today, lasers are widely used in medicine.

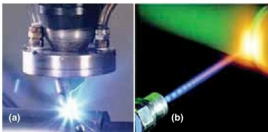


Fig. 12: (a) Laser cutting, (b) laser welding



Fig. 13: Laser eye surgery



Fig. 14: The LAHAT laser-guided missile



Fig. 15: Laser used to increase accuracy of weapons

Lasers are extensively used nowadays in surgery, eye treatment (Fig. 13), dentistry, cancer treatment and other medical operations.

Military applications. There has been a dramatic increase in the commercial, industrial, medical, scientific and military applications of lasers in

the last couple of decades. During this period we have witnessed lasers making their presence felt very effectively in many new areas. While on one hand, the systems used for military applications such as range finders, laser target designators and laser-guided munitions (Fig. 14) continue to improve in terms of performance specifications and system engineering, a lot of effort is being put around the world in the technologically developed countries to exploit the potential of lasers in a domain that is different from that of the existing usage.

Two types of directed-energy weapons are going to play a comprehensive role to enhance both defensive and offensive capabilities of the armed forces of a country having them. These include the so-called low-energy directed-energy weapons and the high-energy directed-energy weapons.

Low-energy laser weapons refer to the electro-optic countermeasures class of laser systems and their different variants. These include anti-sensor laser weapons and anti-personnel laser weapons.

Anti-sensor laser weapons are capable of either temporarily or permanently disabling various kinds of opto-electronic sensors (like PIN photodiodes and CCDs forming the front end of laser range finders, laser tracking systems and seeker heads), thus providing the jamming function.

Anti-personnel laser weapons are capable of again either temporarily or permanently blinding armed personnel, though there seems to be some sort of an agreement among the various countries to develop and deploy only